



COLLEGE
OF
MEDICINE

Intrauterine Fetal Death IUFD

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IUFD

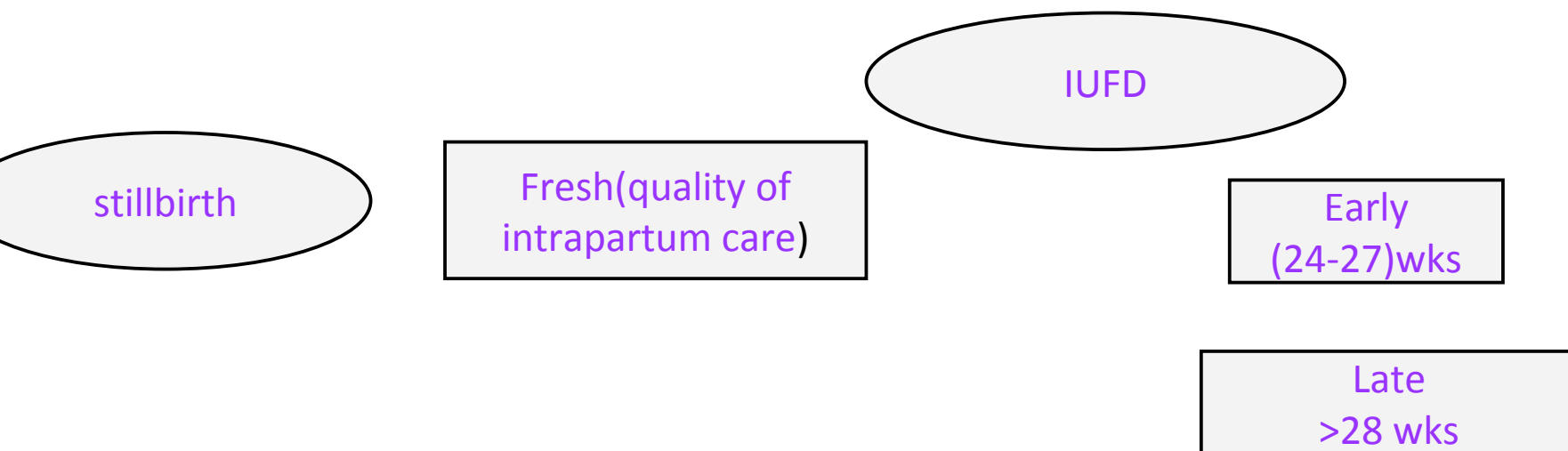
Fetal death before onset of labour or fetus with no sign of life in utero after 24 weeks of gestation

Definition varies Gestational age/ Birth weight

WHO:

Intrauterine fetal death refers to babies with no signs of life in utero after 24 wks of gestation or weighing > 500 grams when gestational age is not known

- Stillbirth: no evidence of life after birth beyond 24 weeks



Incidence and impact

4.5/1000 births

impacts:

Emotionally challenging for :

doctors

parents

increases medicolegal issues

indicator of country health care system

Risk factors and causes

- The RCOG guideline states that parents should be told that no specific causes is identifies in 50% of cases

NO	CAUSES	%
1.	Maternal	5-10
2.	Fetal	25-40
3.	placental	20-35
4.	Unexplained	15-35

Maternal causes (risk factors)

- Obesity ($>30 \text{ kg/m}^2$) : proven ,modifiable
- Maternal age ($>35 \text{ yrs}$)
- Smoking/alcohol/drug abuse
- Infections (hepatitis ,malaria, influenza, toxoplasma ,sepsis)
- Medical disease (DM, HT, Thyroid diseases)
- Autoimmune disorders (APS,SLE)
- Rh incompatibility
- Thrombophilias
- Cholestasis of pregnancy
- Labour related (preterm, dystocia ,uterine rupture)

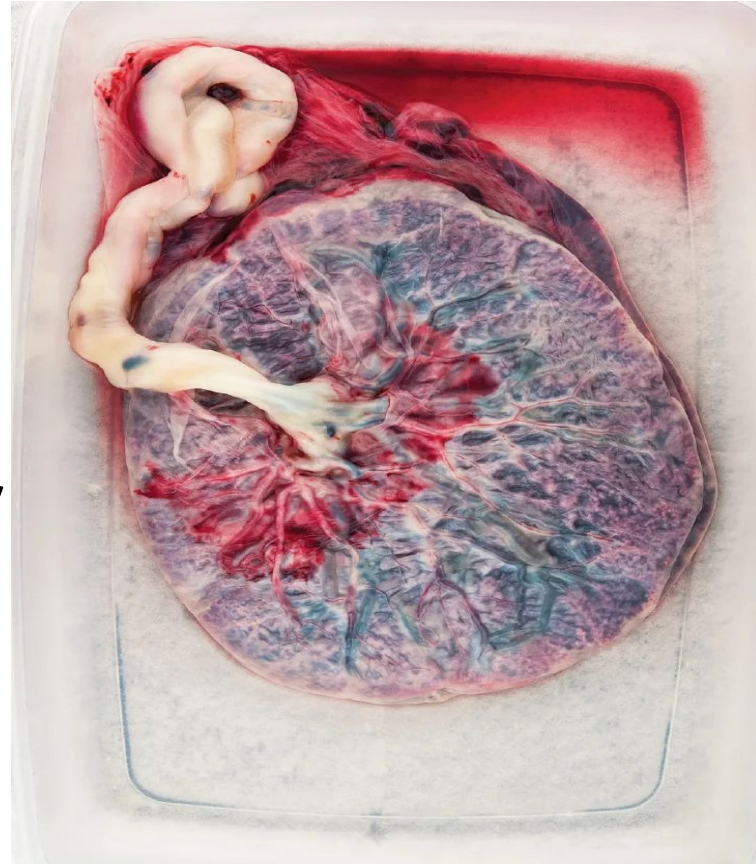
Fetal causes

- Multiple gestation, FETS
- IUGR
- Congenital anomalies
- Infections(GBS, e-coli, klebsiella, rubella ,chlamydia,syphilis, parvovirus ,toxoplasmosis, CMV)
- Hydrops(immune and non-immune)
- G6PD deficiency
- Birth defects
- Prematurity and postmaturity



Placental causes

- Abruptio
- Cord accidents
- Placental insufficiency
- Placenta previa
- Chorioamniotitis
- TTTS
- Feto-maternal hemorrhage Iatrogenic-ECV, drug overdoses





Clinical evaluation

•History

- Last fetal movements
- Symptoms associated with cause (e.g. vaginal bleeding and abdominal pain with abruption, itching with obstetric cholestasis)

•Examination

- No fetal heart heard with sonicaid or CTG.
- Loss of fetal heart during labour with intrapartum deaths
- Examination findings associated with cause (e.g. hard uterus with abruption, hypertension with PE).
- vital signs



Investigations

- Divided into:

- 1) Investigating the cause
- 2) Documenting the death and possible related complications

SYSTEMATIC approach to evaluation

- Varied recommendation based on experts opinion
- Yet, no scientific effective evaluation plan
- Optimal evaluation is must for
 - chance of recurrence
 - future pre-conceptional counselling
 - Pregnancy management
 - plan prenatal diagnostic procedures
 - Neonatal management
- Obvious causes- no further testing or limited testing (cord accidents, anencephaly)

1. History
2. Gross examination
 - SB infant
 - Umbilical cord
 - Placenta
 - Amniotic fluid
3. Fetal autopsy & karyotyping
4. Placental investigations
5. Maternal investigations



1. History

Family

- Recurrent abortions
- Congenital anomalies
- Abnormal karyotype
- Hereditary conditions
- Developmental delay

Maternal

- DM
- HPT
- Thrombophilias
- Autoimmune disease
- Severe Anemia
- Epilepsy
- Consanguinity
- Heart disease

Past Obstetrical

- Baby with congenital anomaly/ Hereditary condition
- IUGR
- Gestational HPT with adverse sequelae
- Placental abruption
- IUFD
- Recurrent abortions

2. *Gross description*

Infant description

Malformation

Skin staining

Degree of maceration

Color-pale, plethoric

Umbilical cord

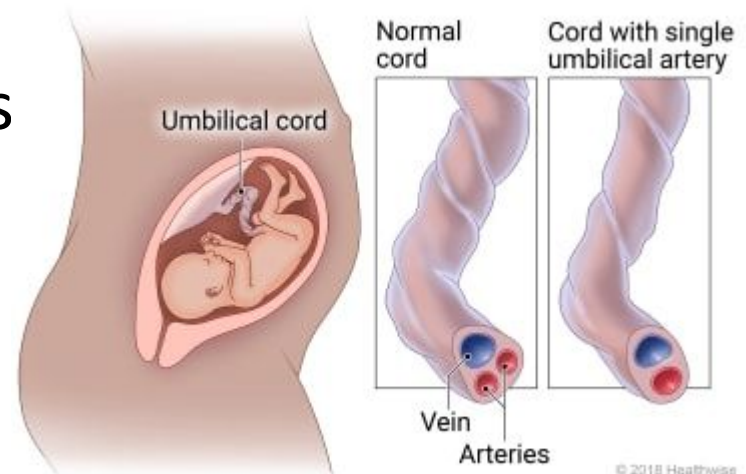
Prolapse

Entanglement-neck, arms, legs

Hematoma or stricture

Number of vessels

Length



Amniotic fluid

Color-meconium, blood

Volume

Placenta

Weight

Staining

Adherent clots

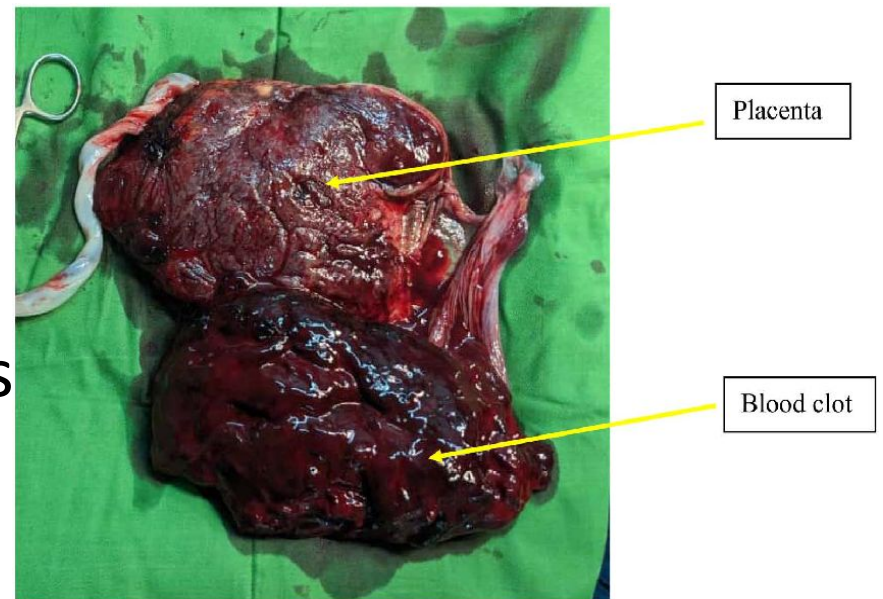
Structural abnormality

Velamentous insertion

Edema/ hydropic changes

Membranes

Stained ,Thickening





3. Fetal autopsy & karyotyping

- These 2 are important tests in SB evaluation
- Crucial for future pregnancy
- Appropriate consent required to take fetal tissue, autopsy
- Ideally should be done by perinatal pathologist
- If denied, post mortem MRI should be considered
- Radiographs if indicated for skeletal abnormalities
- Photographs

- Fetal karyotyping (ACOG recommended in all cases) esp-

- Dysmorphic fetus, FGR
- Hydropic
- Signs of chromosomal anomaly

Samples

- Amniocentesis – highest yield
- 3ml fetal blood from umbilical vs and or cardiac puncture – heparinized bulb
- If blood not obtained ACOG recommends at least 1 of the followings samples-

Pl block 1x1cm, Cord 1.5cm ,Costochondral junction or patella (skin not recommended)

- Parents with multiple pregnancy losses (second or third trimester)
- For aneuploidy- FISH, For small deletions- CGH



4. Placental investigations

- Chorionicity
- Cord knot, vessels, thrombosis
- Infarcts, thrombosis, abruption
- Vascular malformations
- Signs of infection
- Placental block (1x1cm) below cord insertion
- Umbilical segment (1.5cm)
- Placental swabs for infections
- Bacterial cultures for E. Coli, Listeria



5. Maternal evaluation

- CBC
- HB electrophoresis
- Diabetes testing (HbA1c, FBS)
- TFT
- Additional tests

□Kleihauer Betke (for all women, before birth) , in Rh-D negative second test after antidote

□Serological Tests (TORCH, Syphilis, Parvovirus)

?? In all cases, opinion varies, rarely helpful

If clinical findings suggest intrauterine infection (i.e. , those with IUGR, microcephaly)



- Antiphospholipid (LA, ACA) , Antiplatelet Ab if ICH detected
- ?? Thrombophilias screening (6 weeks postpartum) – factor V leiden mutations & deficiencies, antithrombin III, protein C & S
- Bile acids (Cholestasis of preg) – important cause, recurrence in 80% cases
- High vaginal & cervical swab for C & S
- Urine toxicology screening (cocaine, amphetamines are associated with abruption)



Management

- Depends on :

- ☐ Single or multiple gestation

- ☐ Gestation age at death

- ☐ Parents wish (varied response)

- ☐ **Expectant approach**

- 80% goes in labour within 2-3 weeks

- Emotional burden, risk of chorioamnionitis & DIC

- ☐ **Active approach**

Rcog & nice regimen

- <26 weeks – 100 microgram 6hourly (max 4 doses)
- >27 weeks – 25-50 microgram 4hourly (max 6 doses)
- Use of PGs is associated with increase risk of uterine rupture in cases of previous scar
- Membranes should not be ruptured as long as possible
- Pain management should be offered
- Keep watch on CBC, coagulation profile, signs of infection
- Active management of III stage of labour
- Keep blood and blood products ready



Complications

- Infection
- PPH
- Retained placenta
- Abruptio
- DIC
- Shock, renal failure
- Sepsis
- Maternal death



Post delivery

- Emotional support & counseling as they are at increased risk of PPD
- Keep in non-maternity ward
- Suppression of lactation (tight breast support, dopamine agonists, estrogen)
- Counsel for future pregnancy, early ANC visit, preconceptional testing
- Assurance in cases of non recurring causes
- Contraceptive counseling

Management of future pregnancy

- Detailed medical and obstetric history
- Evaluation and workup of previous stillbirth
- Determination of recurrence risk
- Smoking cessation
- Weight loss in obese women (preconception only)
- Genetic counselling if family genetic condition exists
- Medical problem like diabetes should be managed prior
- Thrombophilia workup
- Risk of recurrence
- Support and reassurance



**Thank
You**